KOITA CENTRE FOR DIGITAL HEALTH

NATIONAL CANCER GRID
COLLABORATION FOR CANCER CARE

Health-a-thon 2026

Doctor Knowledge & Productivity Assistant

An assistive, voice-first, source-grounded oncology knowledge co-pilot for research and productivity

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Problem Statement

Oncology knowledge is vast, changes fast and sits in English textbooks and PDFs. Doctors need a checkable answer in the middle of a clinic. Patients and families need the same facts in their own language.

1.56 M

new cancer cases in India in 2024, with about 0.9 M deaths [1]

~1 : 1,500

oncologists to patients in India, against about 1 : 100 in the US [2]

~10 %

of Indians report speaking English; the country has 22 scheduled languages [3]

Doctors lose time

The answer sits in a 4,000-page book. One lung-cancer guideline holds dozens of recommendations that depend on biomarkers, prior drugs and setting.

Patients are locked out

NCCN Guidelines for Patients are reliable but long and in English, and typing is hard for many families.

Chatbots cannot be trusted blindly

General chatbots sound sure but name no source, and can invent facts or references. That is unsafe in cancer care.

Guideline drift

With little time, practice can quietly drift from national standards such as NCCN. Fast, grounded access to the source prevents it.

[1] IARC GLOBOCAN 2024, cited in WHO Global Status Report on Cancer 2026, via ORF (Aug 2026) [2] EY study 2015, via IndiaSpend (dated; estimates vary widely) [3] Census of India 2011

Why a general chatbot is not enough

	Typical general chatbot	This platform
Source of facts	Whatever the model remembers	Only the indexed textbooks, ASCO and NCCN guides
Citations	May cite sources that do not exist	Every sentence cited to book, chapter and page; the page opens in one click
When it does not know	Tends to answer anyway	Says “not available in the sources”, enforced at three points
Treatment recommendations	Generated prose	ASCO statements quoted word for word and ranked by rules
Patient and clinical data	Typed into an online service	English questions stay on our machine; research queries are anonymised before leaving
Language	Mostly English	Voice or text in 11 Indian languages, answer plus English beside it
Model	Large hosted model	Small open-weight Qwen on a CPU, not fine-tuned

The difference is design, not a better prompt. Refusal, citation and verbatim quoting are enforced in code, so a wrong answer is replaced by “not available”.

Target Users and Use Case

Oncologists

Between patients in a clinic or ward

Key needs

- Find a guideline-backed answer fast, with the page to check
- Match a case to the right ASCO recommendation
- Draft handouts and scan the literature

How it fits: Speak the question in English or Hindi; answer with citations in seconds.

Healthcare staff

Nurses, counsellors, coordinators

Key needs

- Look up trusted patient information to explain to families
- Produce patient-education handouts
- Answer routine questions consistently

How it fits: Use it on a phone or laptop with no install or login.

Patients and families

At home or in the waiting room

Key needs

- Understand diagnosis, treatment and side effects
- Ask in their own language, by voice if typing is hard
- Know when to call the care team

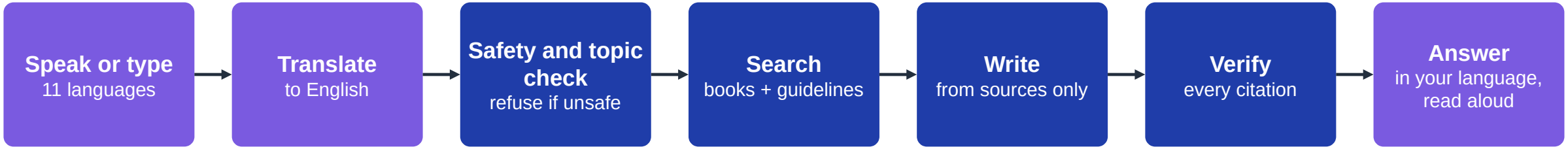
How it fits: Plain-language NCCN answers, read aloud, with a safety screen.

Setting: Indian hospitals, clinics and homes. **Task:** get a trusted, checkable answer or draft quickly, in the user's own language. **Device:** laptop or mobile browser.

Solution Overview: one portal, six tools

A voice-first assistant that answers only from named sources. Doctors and patients speak or type in their own language; the AI searches indexed textbooks and guidelines, writes only from what it finds, checks every citation, and says “not available” when the sources do not cover it.

From question to cited answer



Purple steps use Sarvam AI online; blue steps run on our own machine.

Inputs

- A spoken or typed question
- A case in one sentence
- A short brief for a draft

What the solution does

- Retrieves and ranks passages
- Quotes guideline text, never invents it
- Refuses when evidence is thin

Outputs

- A cited answer, page one click away
- Ranked ASCO recommendations
- A draft flagged for the doctor's review

Key Features: six tools in one portal

Knowledge Assistant

Cited answers from four oncology textbooks, by voice or text. Ask, or Find every place a topic appears.

Local

Treatment Guidance

A case in, matching ASCO recommendations out, quoted word for word and ranked.

Local, no model

ASCO Guidelines

Patient Portal

NCCN patient guides, Ask and Learn, in 11 languages, spoken or typed, read aloud.

Local + Sarvam voice

NCCN National Comprehensive Cancer Network®

Research Platform

Published evidence via scite.ai, anonymised first, summarised locally with [n] citations.

Online: scite.ai

Content Studio

Blogs, handouts and articles from a short brief. Every draft is flagged for review.

Local

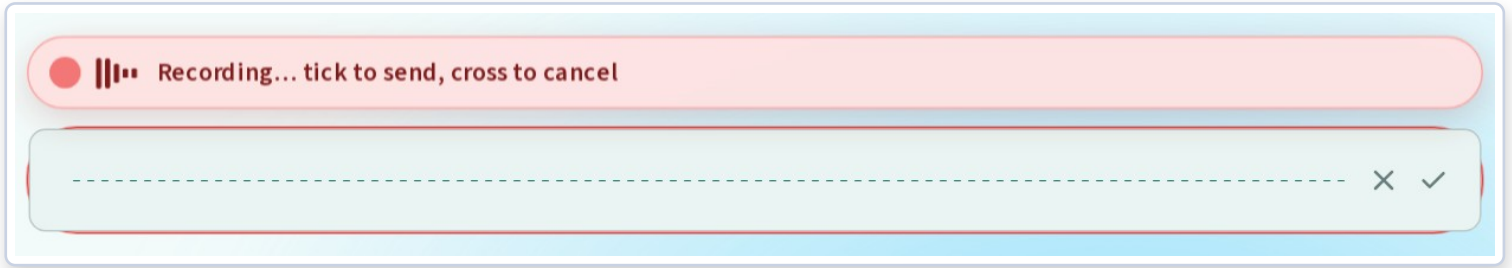
Voice and languages

Speech in, translation at the edges, speech out. The English pipeline is unchanged.

Sarvam AI

Voice-first applies to the Knowledge Assistant (doctors) and the Patient Portal (patients). The next slides show each feature on the live website.

Feature: voice-first for doctors



Press the microphone: a glowing indicator shows it is live

30 s
recording cap, keeps credit use low

EN + HI
English or Hindi/Hinglish for doctors

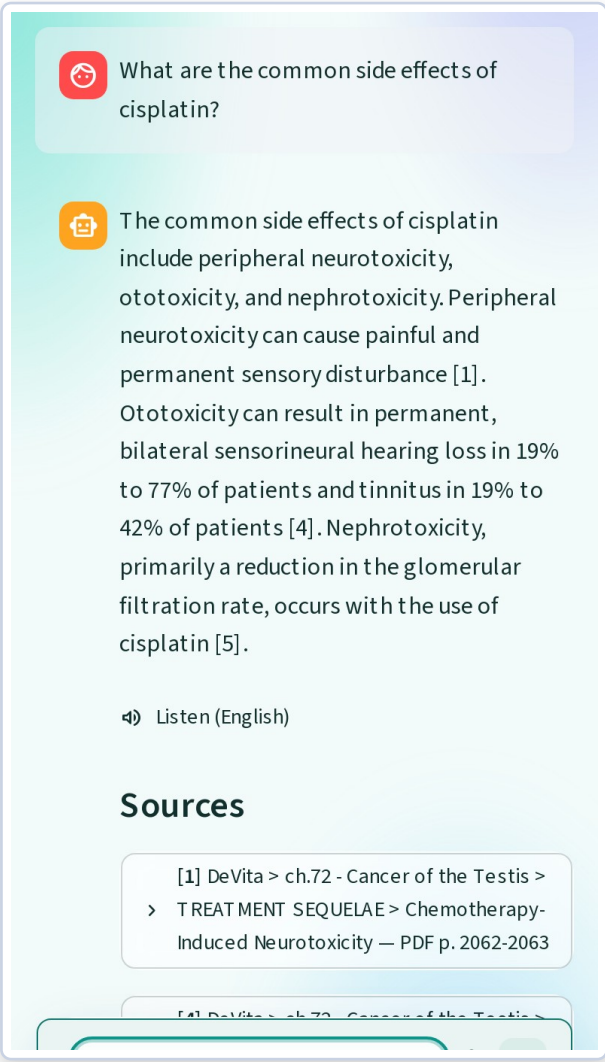
11
languages for patients

Speak, don't type. A doctor between patients just presses the mic.

Drug names, stages and doses stay in English inside Hindi answers.

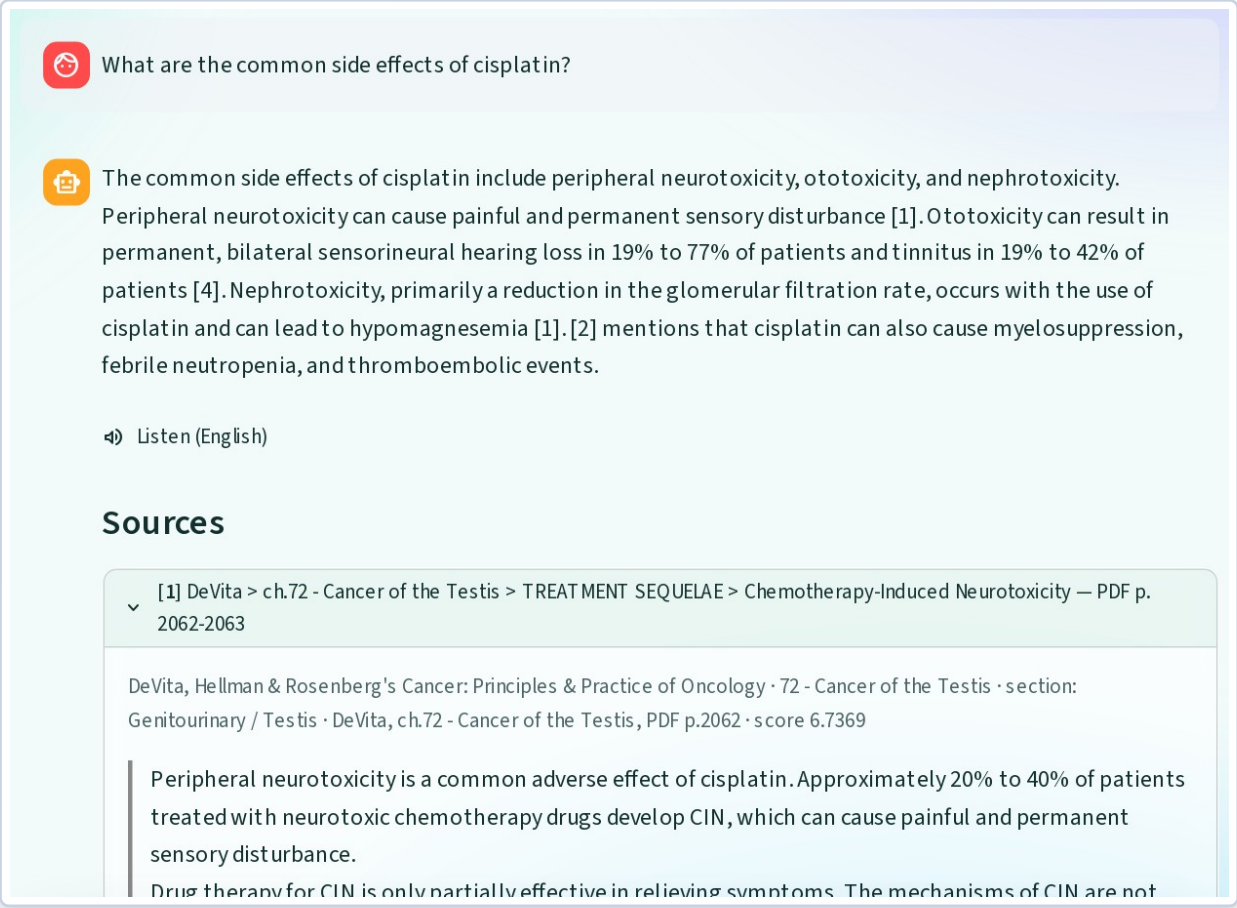
What it heard is shown back, so a mis-hearing is visible before anyone relies on it.

Same pipeline as typing: the same cited answer, or an honest “not available”.



On a phone: spoken question, cited answer

Feature: cited answers from four textbooks



Open any numbered source to read the passage and its page

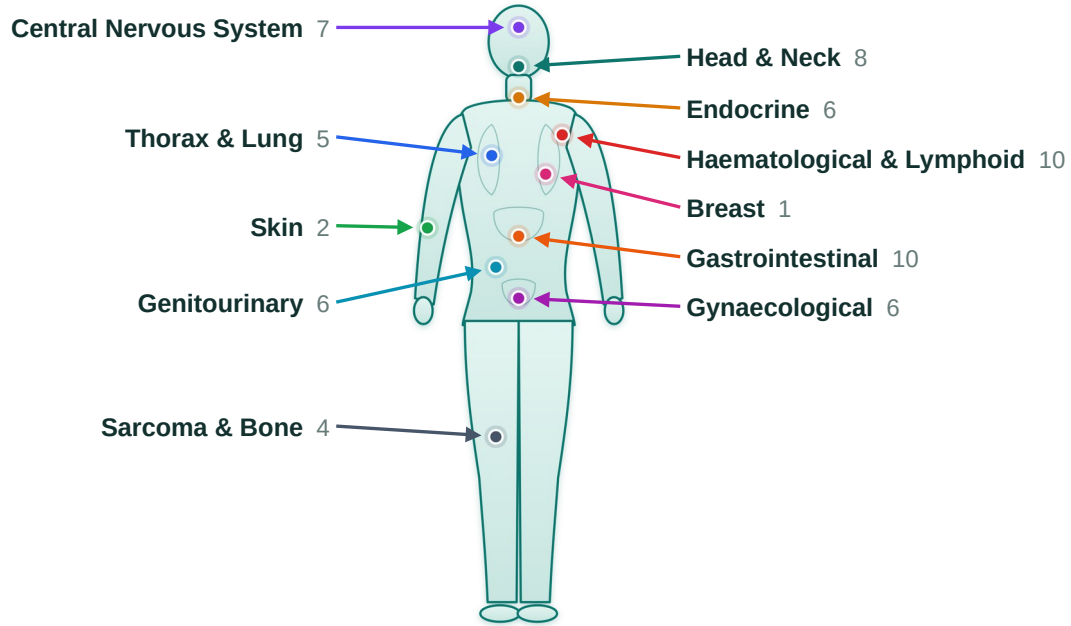
Four textbooks, 19,949 passages: DeVita, Gunderson & Tepper, MD Anderson handbook, Oncologic Imaging.

Every sentence cited to book, chapter and page; the page opens in one click.

Says "not available" when the books do not cover the question: 10/10 off-topic and 8/8 unanswerable questions were refused.

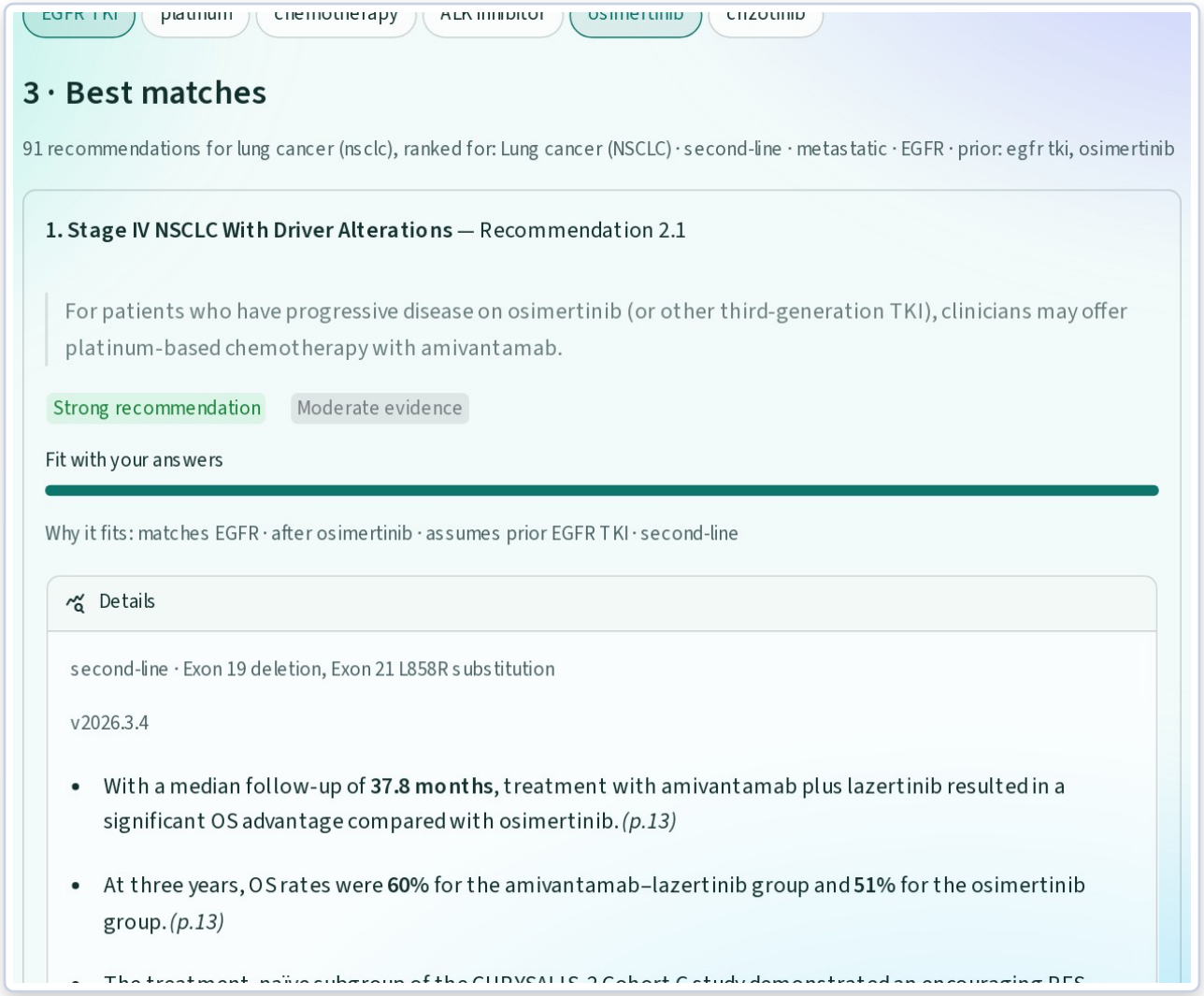
Find a topic: lists where a subject is discussed, page by page, with no model involved.

Filter by organ system or pathology: 14 systems, 78 pathologies



Not site-specific: **Paediatric** 8 | **Cross-cutting** 4 | **General oncology** 1

Feature: Treatment Guidance from ASCO



Case: EGFR-mutant lung cancer progressed on osimertinib

1 Describe the case
Type or paste it, or pick the cancer

2 Confirm the details
Cancer, biomarkers, prior drugs; editable

3 See the best matches
Top results re-ranked after each answer

4 Check the source
Strength, evidence, trial figures, PDF page

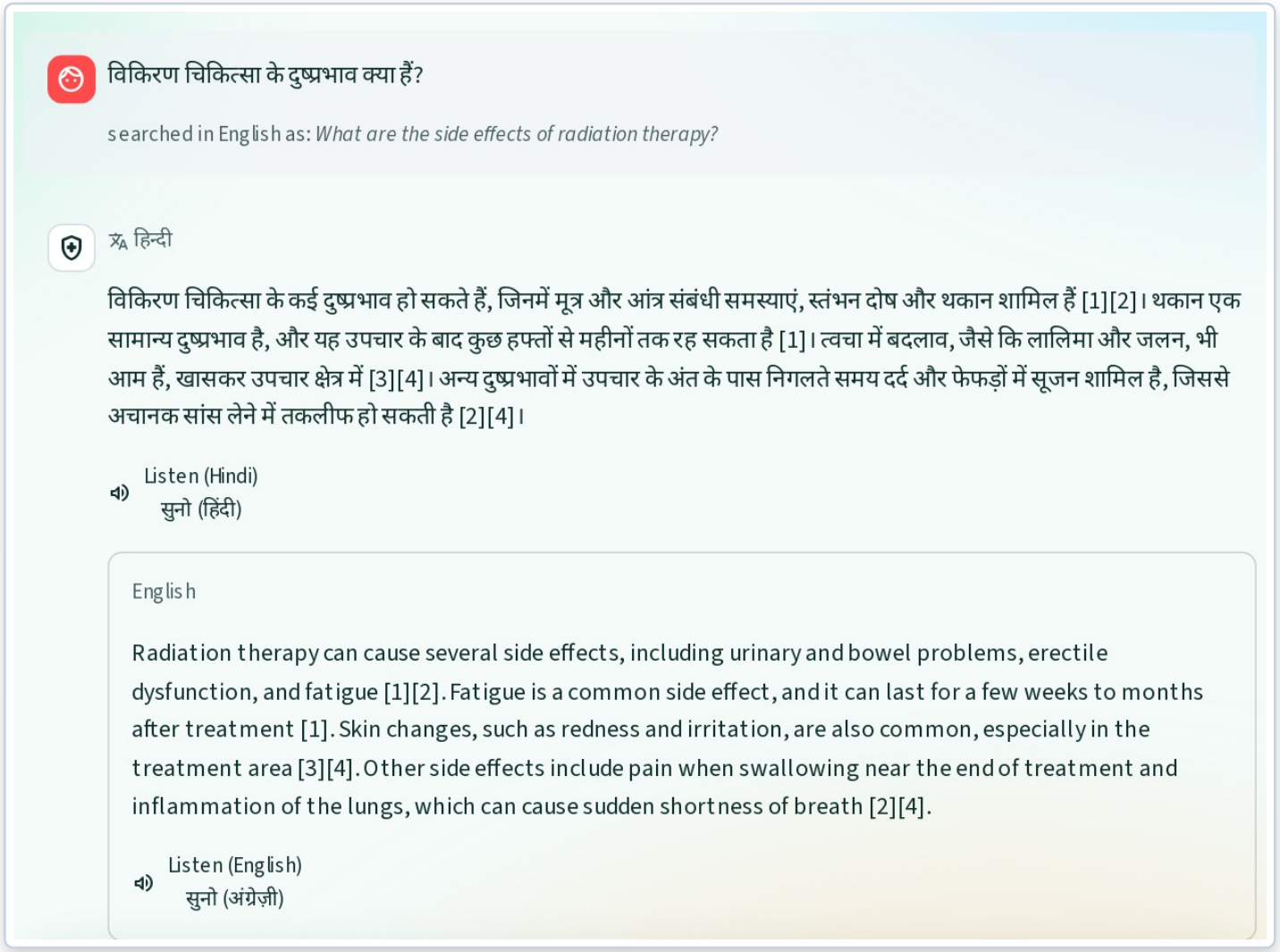
19
ASCO guidelines, 11 cancers

443
recommendations

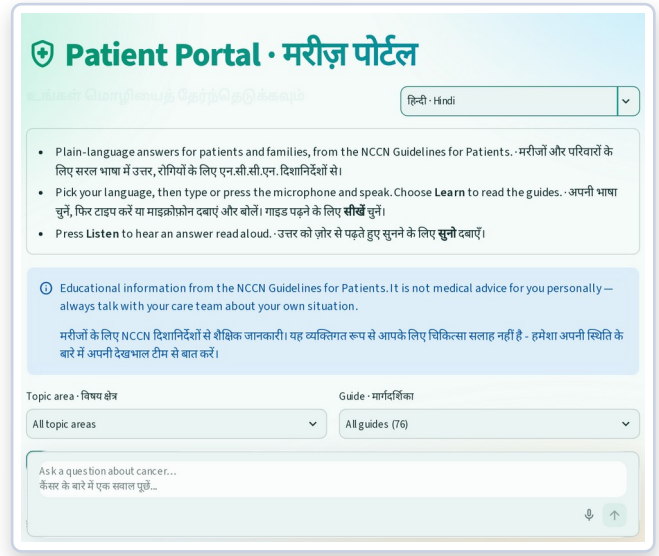
1,264
of 1,264 statements verbatim

It supports the doctor; it does not replace clinical judgment. An uncovered cancer is marked "not covered", and the textbooks are searched as a labelled fallback.

Feature: Patient Portal in 11 languages



A Hindi question: answer in Hindi, English beside it, Listen for each



Pick a language and the page follows (11 available)

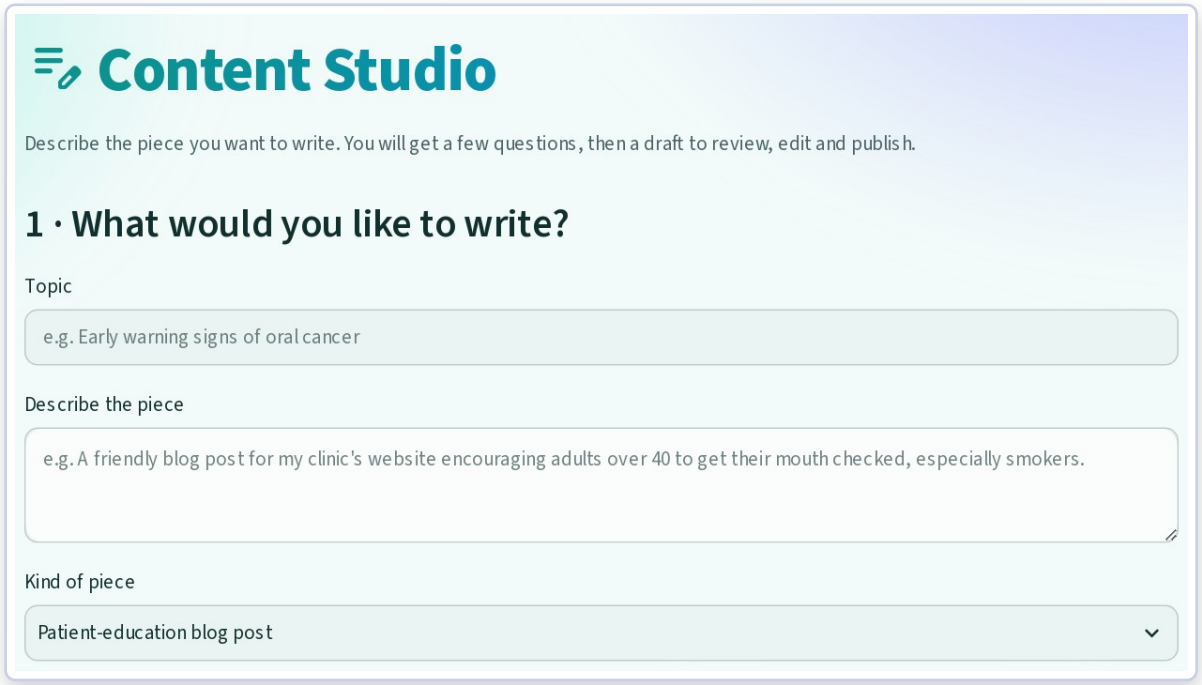
Speak or type in any of 11 Indian languages, or auto-detect.

Plain-language answers from 76 NCCN guides, every sentence cited.

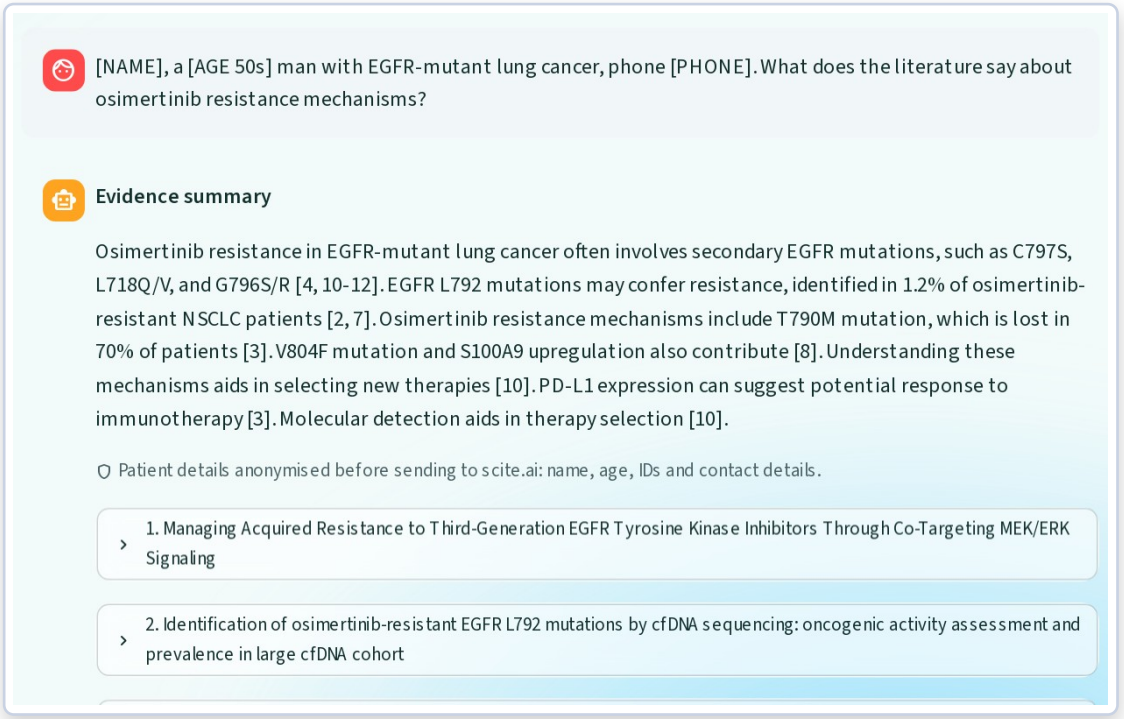
Safety first: crisis words get a helpline, not an answer.

Learn mode: NCCN's own questions, key points and glossary.

Feature: Content Studio and Research Platform



Content Studio: topic, brief, kind of piece



Research Platform: anonymised before it is sent

Content Studio: blog posts, handouts and articles from a short brief and a few questions. Never invents statistics; every draft is flagged “for your review”.

Research Platform: published evidence via scite.ai, summarised locally with [n] citations. Names, age, address and dates are anonymised before anything is sent.

Expected impact

Measured today

~34 s

to a cited answer on a CPU; first words at ~10 s

0 / 35

patient questions wrongly refused

11

Indian languages by voice or text

1,264

ASCO statements all matched word for word

Expected with adoption

Less time searching. One spoken question replaces a hunt across four books and 19 guidelines.

Fewer unsupported answers. Refusal and citation checks stop guesses reaching a clinician or a patient.

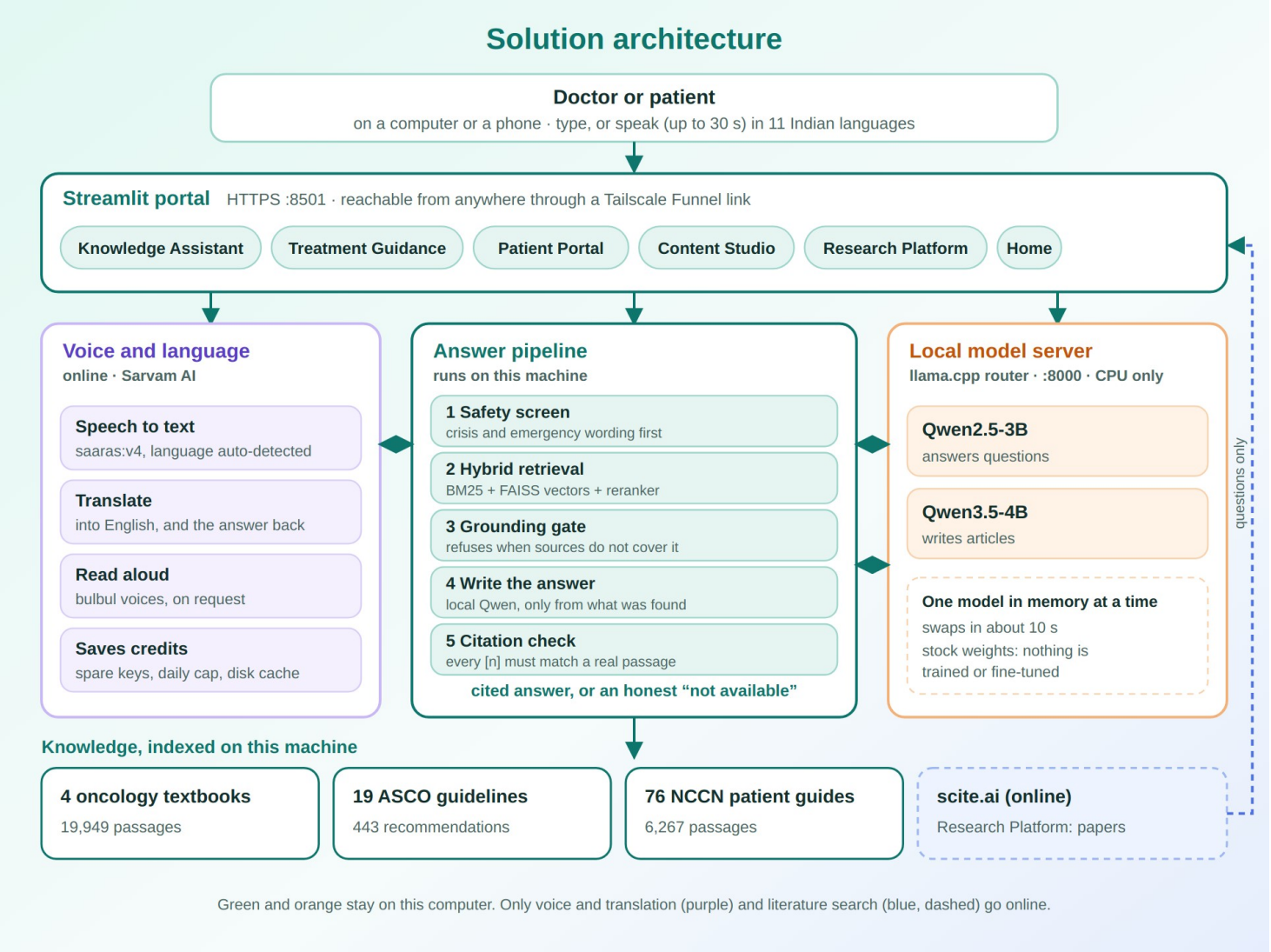
Better access. Patients who cannot read English get NCCN-grade information in their own language.

Less drift from standards because the source page is one click from every answer.

More time with patients as drafting handouts drops from hours to a few minutes.

What we are not claiming. Time saved and patient outcomes are expected, not measured: there has been no controlled user study. The measured figures come from our own labelled test sets and are on the evaluation slides. Drafting a 600-word article takes 2 to 3 minutes on this CPU.

Technology, Tools and AI Approach



How AI is used

Retrieval: keyword + vector search, then a reranker, finds the right passages.

Generation: a small local Qwen writes only from those passages; nothing is fine-tuned.

Verification: rules check every citation and quote; weak evidence is refused.

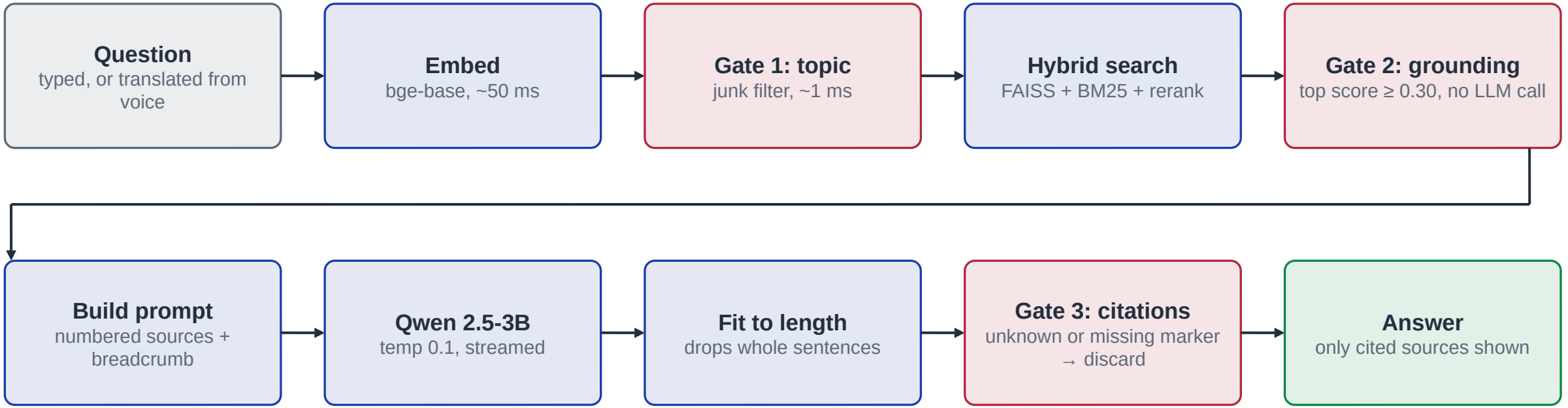
Voice and language: Sarvam AI for speech, translation and read-aloud.

Tools and models

- **Qwen 2.5 / 3.5** — local LLMs
- **llama.cpp** — model server
- **BGE + MiniLM** — embed, rerank
- **Sarvam AI** — voice, translate
- **scite.ai** — literature
- **Streamlit** — web app
- **Python** — backend
- **Tailscale** — public link

Answer pipeline: two gates, one check

Safeguards sit in code, not in a prompt asking the model to behave.



“Not available”
exit at gate 1, 2 or 3: no wrong answer shown

~4 %
of the wait is retrieval; ~96% is the model

34 s
typical total on a CPU, no GPU

Build Sprint Journey and Progress



From the midterm idea to the final product

Evolution since round 1

Midterm: 3 tools, an offline portal, 6,660 pages of textbooks.

Final: 6 tools, 19,949 passages, 19 ASCO guidelines, 76 NCCN patient guides.

New: voice in 11 languages, Treatment Guidance, Patient Portal, public link.

Challenges and what we changed

Small model copied sources. Switched from 1.5B to the 3B model.

Long case vignettes refused. Fails safe; improvement planned.

ASCO matcher got 5/10 on unseen cases. Fixed the gaps it exposed.

scite.ai monthly limit. Falls back to literature search; spare key.

Voice cost. 30 s cap, caching, spare keys, Rs 500/day cap.

Still incomplete or planned

Formal user study with clinicians and patients.

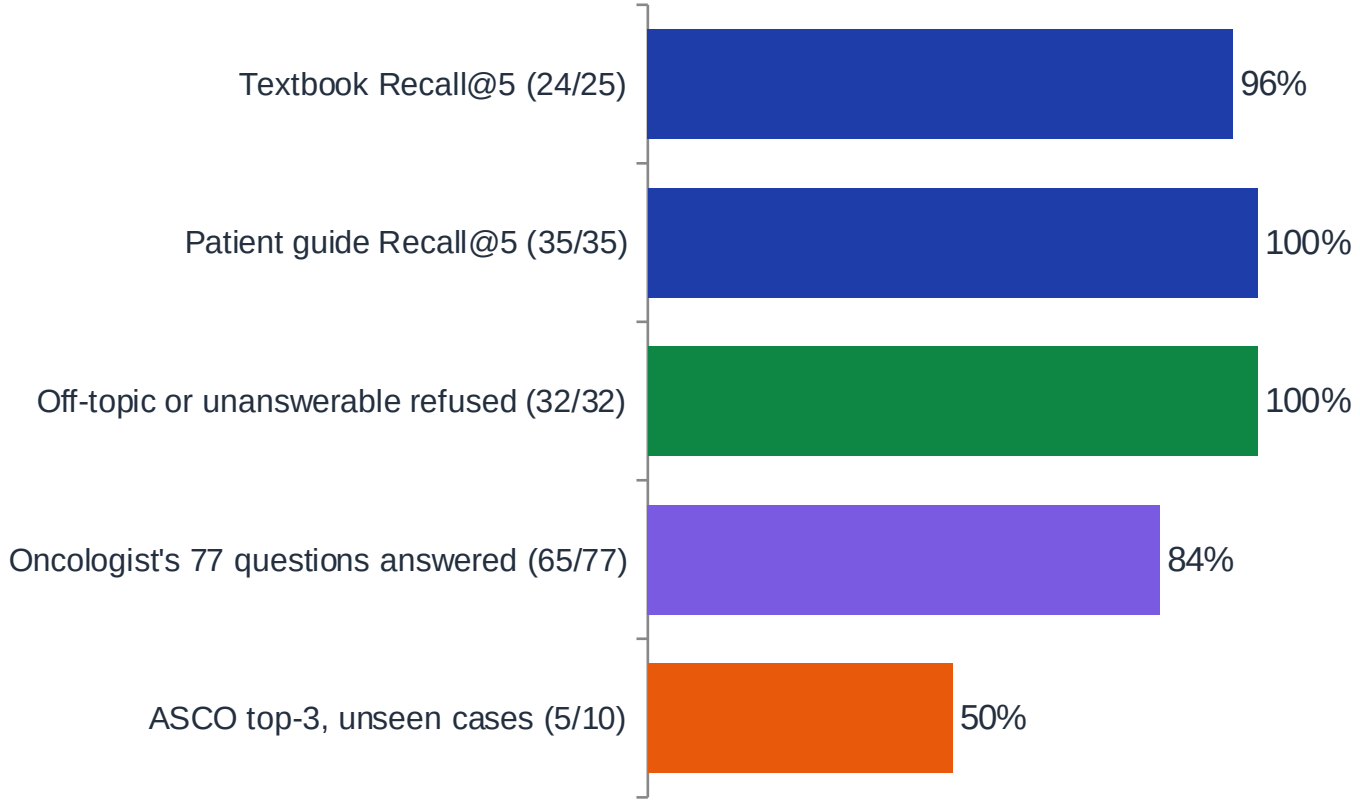
EMR integration and a hospital server.

More specialties beyond oncology.

Same promise, wider reach. The core promise from round 1, answers only from named sources and “not available” otherwise, is unchanged; what grew is who can use it and how.

Evaluation and User Validation: what we measured

Labelled question sets we wrote for each tool, run offline and live. The weak spots are shown too.



High recall. The right passage is in the top five for 96% of textbook questions and 100% of patient questions.

Refuses correctly. Every off-topic and unanswerable question was declined; none got a made-up answer.

Quotes faithfully. All 1,264 ASCO statements match their PDFs word for word.

Honest weak spot. The ASCO matcher put the right recommendation in the top three for 5 of 10 brand-new cases. We report that, not the tuned figure.

Scope: textbook Recall@5 is measured on 25 questions, mostly oral tongue and head-and-neck (page-level ground truth). Other organ systems were checked at organ-system level (38 of 38).

Evaluation detail by tool

Tool	Check	Result
Knowledge Assistant	Right passage in top 5 (Recall@5)	**96% (24/25)**
Knowledge Assistant	Right organ system	**38 / 38**
Knowledge Assistant	Off-topic refused | unanswerable abstained	**10/10 | 8/8**
Knowledge Assistant	77 complex case questions (oncologist)	**65 answered, 12 not in book**
Patient Portal	Right guide page in top 5	**35 / 35**
Patient Portal	Off-topic refused | not-covered refused	**8/8 | 6/6**
Patient Portal	Answerable questions wrongly refused	**0 / 35**
Patient Portal	Safety cases routed correctly	**12 / 12**
Patient Portal	Live answers from guides (3B model)	**10/14, 85% of sentences cited**
Treatment Guidance	ASCO statements matched word for word	**1,264 / 1,264**
Treatment Guidance	Right recommendation in top 3, new cases scored once	**5 / 10**

124 automated pipeline checks and 95 interface checks pass offline, plus live checks with the models running. All figures come from our own labelled sets in the repository; no external benchmark is claimed.

User validation with the oncologist

Dr. Akash Datt, our oncologist partner, supplied 77 complex clinical questions and reviewed the answers. Each was run end to end through the solution.

77
clinical case questions from the oncologist

65
answered with textbook references (84%)

12
declined as “not in book”, never guessed

75 s
average per answer on this CPU (generator Qwen3.5-4B)

Oncologist's review of the answers

Questions: 77 complex clinical questions from our oncologist partner, all run through the assistant and its answers reviewed.

Answered: 65 of 77 (84%) with textbook references, each sentence cited to book, chapter and page.

Declined: 12 of 77 marked “not in book”; the assistant refused rather than guess.

Rated acceptable or correct: *[x of 77: to be added]*

Improvements and limits

Improved: indexed whole books, so in-scope questions are no longer refused (0 of 12 wrongly refused).

Limit: 12 of 77 were declined, several of them long case vignettes the books do cover.

Limit: no formal user study with patients or other clinicians yet.

Safety and Responsible AI

Abstention by design

Three refusal points. A junk query never reaches the model. “Not available” is a first-class answer.

Checkable citations

Every sentence cites book, chapter and page. The page renders in one click.

Quoted, not generated

ASCO and NCCN text is verbatim; a test checks all 1,264 spans against the PDFs.

Patient safety screen

Crisis, emergency and personal-decision patterns are handled before retrieval, in every language. Helpline: Tele-MANAS 14416.

Human in the loop

The doctor confirms case details; drafts are flagged for review; decision support only, not medical advice.

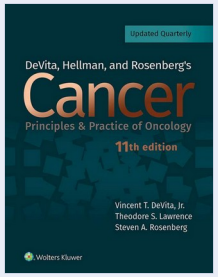
Limits stated plainly

The citation check proves a citation exists, not that it is right. Machine translation is for understanding; English is one click away.

Sources: What Every Answer Is Built On

Six named sources, nothing else. The AI may only write from passages found in these, and every sentence is cited back to them.

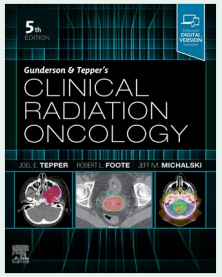
FOUR TEXTBOOKS · 19,949 searchable passages · for clinicians



DeVita, Hellman & Rosenberg's Cancer

Principles & Practice of Oncology, 11th ed. · 4,360 pp

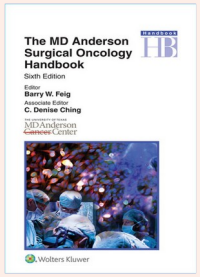
The standard reference: cancer biology, principles of treatment and disease-by-disease management.



Gunderson & Tepper's Clinical Radiation Oncology

2,300 pp

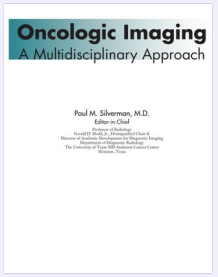
Radiation oncology site by site: indications, techniques, doses and outcomes.



MD Anderson Surgical Oncology Handbook

1,160 pp

A concise surgical guide: work-up, operative approach and follow-up.



Oncologic Imaging: A Multidisciplinary Approach

865 pp

What each cancer looks like on imaging, and how imaging guides staging and decisions.

GUIDELINES · from the societies that write them

ASCO® Guidelines

For clinicians

American Society of Clinical Oncology. 19 guidelines, 11 cancers, 443 recommendations, each with its strength and evidence quality. Powers Treatment Guidance: statements are quoted word for word, never rewritten.

NCCN National Comprehensive Cancer Network®

For patients

National Comprehensive Cancer Network. 76 Guidelines for Patients: expert guidelines in plain language, with key points, questions to ask your doctor and a glossary. Powers the Patient Portal, on a separate index from the textbooks.

Privacy, consent and inclusion

Stays on our machine

- Textbooks, NCCN and ASCO guides, indexes and models
- English Knowledge Assistant, Content Studio, Treatment Guidance and Patient Portal: questions and answers
- No login, no accounts, no patient identifiers requested

Leaves our machine

Research Platform: the question, **anonymised first** (name, age, address, dates, IDs)

Voice: recordings, non-English or translated text and read-aloud text go to Sarvam AI

Transparency: each online step is labelled on screen

Anonymisation guardrail (backend). Before a research question reaches scite.ai, names, ages, addresses, dates and ID numbers are replaced by tags (an age becomes a decade, “54-year-old” → “[AGE 50s]”). It also cleans the answer coming back, and it cannot be switched off.

Bias, accessibility and inclusion

11 Indian languages and voice input for patients who cannot type or read English; phone-ready; plain-language answers; English kept beside every translation.

Known gaps we state

Voice and translation text is not yet anonymised; a bare name with no title or cue is not caught; the public site has no login, so anyone with the link can use Sarvam credit up to the Rs 500/day cap.

Deployment, Cost and Sustainability Plan

Already live. The product is hosted on a free tier and open to patients, doctors and healthcare staff on a laptop or phone, with no install or login.

Free tier

hosting today, over HTTPS

No GPU

one CPU machine, ~8 GB memory

Pay as you go

Sarvam credits, no subscription

Where the cost goes: Sarvam credits

Voice task	Standard price	In our app
Speech to text	Rs 30 / hour	a question: Rs 0.05
Translation	Rs 20 / 10,000 characters	a patient answer Rs 3 to 4
Text to speech	Rs 10 / 10,000 characters	listening to it Rs 1.5 to 2

List prices (sarvam.ai/api-pricing). Cached translations and audio are never billed twice; a 30 s recording cap keeps spend small.

Long-lasting and scalable

Long-lasting: open-weight models and re-indexable sources; a new book or guideline is a rebuild, not a retrain.

Scales up: move to an always-on server (for example an Oracle free ARM VM), then a hospital server with GPU for many users.

Integration: none needed today, it is a browser app; EMR links come later.

What Next? Taking the initiative forward

PHASE 1

Pilot and harden

- Pilot with our clinical partner and a partner hospital
- Anonymise the voice and translation path
- Run a formal clinician and patient user study

PHASE 2

Deepen the product

- Better handling of long case descriptions
- Stronger citation fact-checking
- More languages and more guidelines

PHASE 3

Scale

- Multi-specialty indexing beyond oncology
- Hospital server with GPU for many users
- EMR integration

YES

We are happy for all solution information to be made publicly available, for example on Health-a-thon or other websites. The source books and guidelines are copyrighted and stay out of the public code; anyone can rebuild with their own copies.

Health-a-thon 2026

Thank you

Team Arjun • A5YUIE Doctor Knowledge & Productivity Assistant

TEAM

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LINKS

MVP LINK

v.taila0f4fa.ts.net

DEMO VIDEO

Watch video

CODE REPOSITORY

github.com/viren711/healthathon